

VEDANT RALEKAR

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Summary

Backend-focused Software Engineering student (Final year, EXTC with AI/ML minor) with hands-on experience building production-style RESTful APIs, caching layers, and authentication systems in Node.js/Express, plus applied ML pipelines (RAG, vector search). Solved 500+ DSA problems across LeetCode, GeeksforGeeks, Codeforces and CodeChef. Seeking an SDE/Backend Engineering internship or full-time role to build scalable, reliable systems.

Technical Skills

Programmings: C++, Python, SQL, JavaScript

Backend: NodeJs, ExpressJs, REST APIs, Jwt Authentication, Bcrypt, Microservices, Socket.io, Multer

Database/Query Languages: MySQL, MongoDB, Redis, Firebase

AIML/Data Analysis: RAG Pipeline, Langchain, ChromaDB, Sentence Transformer, Numpy, Pandas, Matplotlib

VCs/Tools: Git, GitHub, CI/CD Pipeline, GitHub Actions, Postman, Render, Docker, Android Studio

CS Core: Data Structures and Algorithms, OOPs, Operating System, DBMS

Projects

AskDSA - RAG Search Engine | FastAPI, Python, Node.Js, Express.Js, Gemini API

[Code](#)

- Exposed a Retrieval-Augmented Generation pipeline through a **FastAPI** backend, integrating **Google Gemini** to power an interactive AI tutoring assistant for DSA topics.
- Combined BM25 keyword search with vector search using Reciprocal Rank Fusion(RRF), MMR, and Cross Encoder reranking, cutting hallucinations approximately by **90%**.
- Engineered **ChromaDB** infrastructure with **20K+ embeddings** and metadata-enhanced retrieval.
- Optimized HuggingFace sentence-transformer embeddings with normalization to improve semantic retrieval quality and **MMR-based RAG** context relevance.

URL Shortener | NodeJs, Express.Js, MongoDB, Redis, Docker, Nginx

[Code](#)

- Designed and built **RESTful APIs** and integrated **Redis** caching, achieving **sub-20ms** URL redirection latency.
- Implemented secure Signup/Login using JWT authentication, **bcrypt** hashing, and **cookie-based sessions**.
- Implemented an **IP-based** rate limiter to throttle abusive traffic and protect API endpoints from **DoS/DDoS**.
- Containerized the full application with **Docker Compose** for reproducible builds and streamlined local/CI testing.
- Reduced database load by approx **60%** and achieved 2x faster redirects by introducing Redis caching layer.
- Engineered horizontal scaling with Docker (6 replicated containers) behind an **Nginx** load balancer, handling **1M+ HTTP** requests at 1K concurrency with **100%** successful responses.

Enhanced Caesar Cipher Tool | C++ Core project

[Code](#)

- Built a high-performance C++ **Caesar Cipher** tool supporting **text and file encryption/decryption**.
- Optimized character transformations with **automatic key normalization**, achieving **90%** faster processing.
- Added **file input/output support** for large file encryption/decryption with **robust error handling**.

Education

Shri Guru Gobind Singji Institute of Engineering and Technology, Nanded

2023 - 2027

B-tech, Electronics And Telecommunication Engineering Minor in AI/ML

8.29/10 CGPA

L.B.U Jr College, Wardha

2022 - 2023

Higher Secondary Certificate(HSC) Board

81.83/100 Percent

Competitive Programming and Publication

- CodeForces:** 700+ rating, solved 70+ problems | **CodeChef:** 1200+ rating, solved 50+ problems.
- Solved **500+ problems** across platforms including **LeetCode**, **GeeksforGeeks**, **Codeforces**, and **CodeChef**.
- Published a research paper in **IJRAW** on *Gen-Next Attendance System using RFID*. [Link](#)

Position/Responsibilities

Core Committi Member-ELITE

- Served as a Core Committee Member, contributing to the planning and execution of college-level events.
- Coordinated with faculty, volunteers, and participants to ensure smooth event operations.
- Assisted in decision-making and problem-solving during event execution.
- Developed leadership, communication, and organizational skills through active involvement.

Certifications [Link](#)

- SQL using MySQL and Database Design - from Scaler Topic.